Fiscal Year:	FY 2013	**Task Last Updated:**	FY 03/05/2013
PI Name:	Zanello, Susana Ph.D.		
Project Title:	Hindlimb suspension (HS) as an analog model of ocular alterations associated with cephalad fluid shifts: resveratrol as a countermeasure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	KBR/NASA Johnson Space Center		
PI Address 1:	Human Research Program Chief Scientist Office		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:	NOTE (January 2021): PI now at KBR/NASA JSC as of December 2020. Previously at imec USA from June 2019-November 2020; NASA JSC (KBRwyle) from August 2017 until spring 2019. Prior to August 2017, PI was with Universities Space Research Association.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/25/2011	**End Date:**	10/01/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:	Shuttle		
Flight Assignment:	STS-133 ; STS-135 NOTE: Gap changes per IRP Rev E (Ed., 3/25/14) NOTE: End date changed to 10/1/2012 to accommodate final reporting since HRP MTL dated 12/28/12 showed 9/30/2012 end date (Ed., 3/5/2013) NOTE: end date changed to 9/13/2012 (previously 9/30/2012); title changed to "Hindlimb suspension (HS) as an analog model of ocular alterations associated with cephalad fluid shifts: resveratrol as a countermeasure" (previously "Part I-Microgravity effects on the mouse retina: Histological and gene expression changes after flight on STS-133. Part II-Hindlimb suspension (HS) as an analog model of ocular alterations associated with cephalad fluid shifts: resveratrol as a countermeasure")--changes per HRP SMO information dated 11/11/2011 (Ed., 12/14/2011)		
Key Personnel Changes/Previous PI:	Additional technical assistance: Corey Theriot, PhD (nucleic acids isolations); Chris Westby. PhD (enzyme immunoanalysis); James Fiedler, PhD (biostatistics); Aziza Glass (summer intern); Audrey Nguyen (CEP-USRA intern)		
COI Name (Institution):	Boyle, Richard (NASA Ames Research Center) Alway, Stephen (West Virginia University School of Medicine)		

Grant/Contract No.:	Directed Research
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Visual function and neuroanatomical ocular changes have been observed in some long-duration crewmembers. These alterations seem to be related to microgravity-induced fluid shifts and ensued ICP increase. The mechanisms by which these factors cause eye damage are not understood. Damage may extend beyond anatomical alterations of the eye. The hypothesis that these two pilot studies will test is that pathophysiological pathways at the molecular and cellular levels in the retina may also contribute to microgravity-induced vision risks. These pathways may be triggered by oxygenation perturbations and mechanical and oxidative stress associated to the fluid shifts in microgravity. In Part I, a mouse experiment aboard two shuttle missions, STS-133 and STS-135, will allow us to test the hypothesis that mouse eyes in microgravity (together with confounding space environment factors) exhibit alterations in the histological and gene expression profile that may be indicative of cellular stress pathways. To date, there have been limited flight animal studies aimed at studying the effects of microgravity on the retina (Tombran-Tink, Adv Exp Med Biol, 2006; Grigoryan, Adv Space Res, 2002). In Part II, a ground animal analog, the rat hindlimb suspension model (HS), will be used to test the same hypothesis using a model of simulated microgravity. A second hypothesis within Part II will test the efficacy of nutritional countermeasures. Resveratrol, a potent antioxidant and activator of the cellular stress response pathway, has been shown to ameliorate bone loss (Habold et al, J Bone Min Res 2010) and muscle atrophy in hindlimb suspended animals (Jackson et al, Am J Physiol Regul Integr Comp Physiol 2010). Likewise, resveratrol has been shown to prevent and ameliorate oxidative damage and retinal neurodegeneration in several experimentally induced ocular conditions in animals (Liu et al, IOVS 2007; Shindler et al, IOVS 2007; Kubota et al, IOVS 2009). This study will examine whether resveratrol has a cytoprotective effect in the retinas of rats subjected to the stress caused by cephalad fluid shifts. In addition, the protective effects of resveratrol will be compared to those of another reactive oxygen species scavenger, epigallocatechin gallate (EGCG) found in green tea extract. This project aims at testing the overall hypothesis that cephalad fluid shifts elicited by either microgravity or hindlimb suspension (HS) represent a stress factor that induces optic disc neuroanatomical changes, as well as retinal cell deterioration and loss via oxidative stress. The first specific aim of this pilot project is to perform histological and gene expression analysis of retinas collected from mice flown in STS-133 and STS-135 and from their ground control counterparts. The second specific aim is to perform similar analysis in retinas from hindlimb suspended rats, focusing on pathways consistent with cellular and oxidative stress, thus validating this ground analog animal model for the study of retinal stress in microgravity. The third specific aim will explore the potential efficacy of reversatrol and EGCG as a nutritional countermeasures to mitigate the risks or retinal oxidative stress and degeneration in conditions of simulated microgravity. These experiments constitute tissue sharing endeavors among collaborators within and also external to NASA. Dr. Zanello leads the JSC efforts and has collected the eyes of the animals from both of these experiments in order to be used for histological and gene expression analysis. References Grigoryan EN, Mitashov VI, Anton HJ. Urodelean amphibians in studies on microgravity: effects upon organ and tissue regeneration. Adv Space Res. 2002; 30(4): 757-64. Habold C, Momken I, Ouadi A, Bekaert V, Brasse D. Effect of prior treatment with resveratrol on density and structure of rat long bones under tail-suspension. J Bone Miner Metab. 2011; 29(1): 15-22. Jackson JR, Ryan MJ, Hao Y, Alway SE. Mediation of endogenous antioxidant enzymes and apoptotic signaling by resveratrol following muscle disuse in the gastrocnemius muscles of young and old rats. Am J Physiol Regul Integr Comp Physiol. 2010; 299(6): R1572-81. Kubota S, Kurihara T, Mochimaru H, Satofuka S, Noda K, Ozawa Y, et al. Prevention of ocular inflammation in endotoxin-induced uveitis with resveratrol by inhibiting oxidative damage and nuclear factor-kappaB activation. Invest Ophthalmol Vis Sci. 2009; 50(7): 3512-9. Liu Q, Ju WK, Crowston JG, Xie F, Perry G, Smith MA, et al. Oxidative stress is an early event in hydrostatic pressure induced retinal ganglion cell damage. Invest Ophthalmol Vis Sci. 2007; 48(10): 4580-9. Shindler KS, Ventura E, Rex TS, Elliott P, Rostami A. SIRT1 activation confers neuroprotection in experimental optic neuritis. Invest Ophthalmol Vis Sci. 2007; 48(8): 3602-9. Tombran-Tink J, Barnstable CJ. Space flight environment induces degeneration in the retina of rat neonates. Adv Exp Med Biol. 2006; 572: 417-24.
Rationale for HRP Directed Research:	This research is directed because NASA must define complete scientific activities in a short time and there is insufficient time to issue a solicitation.
Research Impact/Earth Benefits:	The ocular perturbations found in some astronauts returning from long duration missions resemble those found in high altitude retinopathy and papilledema associated with idiopathic intracranial hypertension (IIH or pseudotumor cerebri). Although no definitive etiology has been proven, there are several biologically plausible hypotheses including transient venous hypertension in the brain and eye secondary to the cephalad fluid shifts with concomitant choroidal volume and IOP changes. An animal model replicating this cephalad fluid shift might shed light upon astronaut related optic disc edema as well as, possibly, terrestrial IIH. Increased knowledge and developed concepts will add value to the understanding of age-related ophthalmic indications on Earth, and possible applicable nutritional countermeasures.

Task Progress:	Upon completion of this task, we have expanded the immunohistopathologic analysis of the rodent retina as far as the effects of and responses to spaceflight observed in the eyes of mice aboard shuttle mission STS-133 and STS-135, focusing, for the first time, on molecular and cellular processes. While the results reported here represent pilot data due to the small sample size, these data provide for the first time, direct evidence suggesting that oxidative stress, neuronal damage, and mechanical injury take place in the retina and optic nerve of rodents flown in low-Earth orbit for a period under 2 weeks. Our work gives a first insight into the impact of space-associated factors on biological processes like cell death, oxidative stress, and probable mechanical injury in the rodent eye. Moreover, comparative results obtained with two different mouse strains (albino versus pigmented mice) suggest that susceptibility to stress factors is determinant of the extent of the stressors effects. While individual variability is not regularly encountered with genetically homogenous mouse strains, we observed some individual susceptibility variation, which may be ascribed to a particular stress impact in spaceflight, part of which is stochastic in nature (radiation). The present work also investigated the contribution of cephalad fluid shift alone in the rat retina by using the well-characterized hindlimb suspension (HS) model, and identified, for the first time, a gene expression response suggestive of mechanical stress imposed by cephalad fluid shift on the eye. In addition, an antioxidant nutritional countermeasure based on a green-tea enriched diet resulted in a decreased stress response, suggesting an association between the possible mechanical stress and oxidative stress genetic circuits.
Bibliography Type:	Description: (Last Updated: 09/04/2023)
Articles in Peer-reviewed Journals	Poritz JM, Seaton KA, Zanello SB, Cromwell RL. "Outcome measure to assess head-down bed rest subject performance." Aviat Space Environ Med. 2012 Jul;83(7):691-5. PubMed PMID: 22779313 , Jul-2012
Articles in Peer-reviewed Journals	Taibbi G, Kaplowitz K, Cromwell RL, Godley BF, Zanello SB, Vizzeri G. "Effects of 30-day head-down bed rest on ocular structures and visual function in a healthy subject." Aviat Space Environ Med. 2013 Feb;84(2):148-54. PubMed PMID: 23447853 , Feb-2013
NASA Technical Documents	Alexander DJ, Gibson CR, Hamilton DR, Lee SMC, Mader TH, Otto C, Oubre CM, Pass AF, Platts SH, Scott JM, Smith SM, Stenger MB, Westby CM, Zanello SB. "Evidence Report: Risk of Spaceflight-Induced Intracranial Hypertension and Vision Alterations." Evidence Book for HRP Roadmap, 2012. http://humanresearchroadmap.nasa.gov/Evidence/ , Jul-2012
Papers from Meeting Proceedings	Cromwell RL, Zanello SB, Yarbough PO, Taibbi G, Vizzeri G. "Comparison of ocular outcomes across two 14-day bed rest platforms." International Society for Gravitational Physiology meeting, Aberdeen, Scotland, June 2012. International Society for Gravitational Physiology meeting, Aberdeen, Scotland, June 2012. ISGP Proceedings. In press as of February 2013. , Feb-2013